

Sankhadeep Bera

Kolkata, India

berasankhadeep20@gmail.com

| [berasankhadeep20-lang.github.io](https://github.com/berasankhadeep20)

| [GitHub](#)

| [LinkedIn](#)

| [Codeforces: Ronnie_Deep_04](#)

Education

Indian Institute of Science Education and Research (IISER) Kolkata

2025 – Present

BS-MS Dual Degree Program | *Physics with focus on ML, Simulation & Data Science*

Bhavans Gangabux Kanoria Vidyamandir

2025

Class XII (CBSE) – 89.8%

St. Joseph's College, Bowbazar

2023

Class X (ICSE) – 94%

Competitive Examination Performance

- JEE Main 2025 – **94.89 Percentile**
- WBJEE 2025 – **Rank 1596**
- IISER Aptitude Test (IAT) – **Rank 3065**

Projects

F1 Pitwall Dashboard

2026

React 18, Vite, Tailwind CSS, OpenF1 API, Recharts, Framer Motion

- Integrated OpenF1 real-time telemetry endpoints to analyse live sector times, DRS activation patterns, tyre age, and inter-car gaps across a full race weekend.
- Built tyre degradation visualisation, lap pace comparison charts, and a championship probability model updating dynamically during live sessions.
- Implemented broadcast mode, driver telemetry overlays, and race strategy simulation with pit stop window analysis.
- Deployed via GitHub Actions CI/CD to GitHub Pages with automated build pipeline.
- Repository: <https://github.com/berasankhadeep20-lang/f1-pitwall-dashboard>

F1 AI Race Outcome Predictor

2026

Python, Scikit-learn, FastF1, Pandas, Plotly

- Built an ML pipeline ingesting multi-season FastF1 telemetry data – historical lap times, driver performance metrics, and circuit-specific statistics – to predict race finishing positions.
- Engineered features from raw telemetry including sector time deltas, tyre compound history, and grid position impact scores.
- Developed an interactive visualisation dashboard displaying model predictions alongside real session data for direct comparison.
- Repository: <https://github.com/berasankhadeep20-lang/F1-AI-Predictor>

Stock Market ML Dashboard

2026

Python, React 18, Vite, Finnhub API, Recharts, GitHub Actions

- Built a production-ready financial analytics platform with live price simulation via Finnhub API, LSTM-style predictive modelling, and full suite of technical indicators (RSI, MACD, Bollinger Bands).
- Architected a real-time data pipeline processing streaming price data with sub-second UI updates via WebSocket integration.
- Deployed automatically to GitHub Pages via GitHub Actions with environment-variable-secured API key management.
- Repository: <https://github.com/berasankhadeep20-lang/Stock-market-ml>

IPL Auction Simulator (Multiplayer)

2026

React 18, TypeScript, Vite, Firebase Realtime Database, Web Speech API, GitHub Actions

- Built a full-stack real-time multiplayer IPL auction game supporting 1–8 human players with AI-controlled teams filling remaining slots, using 500+ real player dataset.
- Implemented Firebase Realtime Database with atomic transactions to prevent race conditions during concurrent bids across distributed clients.
- Integrated Web Speech API for live text-to-speech auction commentary with per-team budget tracking and live leaderboard.
- Repository: https://github.com/berasankhadeep20-lang/IPL_Auction_Multiplayer_Internet

MATCHDAY Football Dashboard

2026

React 18, Vite, Tailwind CSS, football-data.org API, Cloudflare Workers

- Developed a live football analytics dashboard consuming the football-data.org API for real-time match scores, standings, and fixture data across multiple leagues.
- Built and deployed a Cloudflare Worker as a CORS proxy to securely route API requests without exposing keys client-side.
- Deployed as a sub-path SPA to GitHub Pages with Vite base-path configuration and GitHub Actions automation.
- Repository: <https://github.com/berasankhadeep20-lang/matchday-dashboard>

AARSHI Official Website

2026

React 18, Vite, Tailwind CSS, Firebase Authentication, Firestore

- Architected and built the full SPA for AARSHI, IISER Kolkata’s dramatics society, with Firebase-backed member portal, authentication, and event management system.
- Implemented dynamic event modals, member registration flow, and role-based access control via Firestore rules.
- Deployed to GitHub Pages with CI/CD pipeline; maintained strict content integrity policy across the production site.
- Live: <https://aarshiofficial.github.io>

Global Freight Rate Tracker

2025

Python, yfinance, FRED API, SQLite, Plotly, GitHub Actions

- Automated data pipeline fetching freight rate indices via yfinance and FRED API, storing results in SQLite with Z-score anomaly detection and Pearson correlation analysis.
- Auto-deploys an interactive Plotly dashboard to GitHub Pages every 6 hours via scheduled GitHub Actions workflow.
- Repository: <https://github.com/berasankhadeep20-lang/freight-rate-tracker>

AI Football Match Outcome Predictor

2025

Python, Scikit-learn, Pandas

- Trained a classification model predicting football match outcomes (win/draw/loss) from multi-season historical team statistics with feature engineering and cross-validated preprocessing.
- Applied feature importance analysis to identify key performance indicators driving prediction confidence.
- Repository: <https://github.com/berasankhadeep20-lang/AI-Football-Match-Outcome-Predictor>

LLM for Stock Market Analysis

2025

Python, NLP, LLM APIs, Machine Learning

- Developed an LLM-based question-answering system for stock market queries, combining financial text analysis with structured data retrieval for predictive insights.
- Integrated retrieval-augmented generation (RAG) patterns to ground model responses in live financial context.
- Repository: <https://github.com/berasankhadeep20-lang/LLM-For-stock-market>

JARVIS – AI Desktop Assistant

2025

Python, Tkinter, SpeechRecognition, Google Calendar API

- Built a modular voice-controlled desktop assistant with speech recognition, file management, and Google Calendar integration using background threading for non-blocking operations.
- Designed a custom dark-theme Tkinter GUI with persistent session state and command history.
- Repository: <https://github.com/berasankhadeep20-lang/Jarvis-A.I>

Technical Skills

Languages: Python, TypeScript, JavaScript, Java

Web / Full-Stack: React 18, Vite, Tailwind CSS, HTML, CSS

Databases & Cloud: Firebase Realtime Database, Firestore, SQLite

ML / Data: Scikit-learn, Pandas, Plotly, Recharts, FastF1, NumPy

Motorsport: FastF1, OpenF1 API, telemetry analysis, race strategy modelling, tyre degradation analysis

Tools: Git, GitHub Actions (CI/CD), Linux, Cloudflare Workers, WebSocket, DaVinci Resolve

Competitive Programming: Codeforces (handle: Ronnie_Deep_04)

Fields of Interest: Motorsport Data Analytics, Machine Learning, Sports Analytics, Quantum Computing, CFD Simulation

Workshops & Certifications

- **Julia Programming Workshop** – Slashdot, Coding Club of IISER Kolkata
- **Quantum Computing with Qiskit** – IBM [Certificate]
- **Colour Grading using DaVinci Resolve** – Pixel Club & Slashdot, IISER Kolkata

Positions of Responsibility

- **Active Member** – AARSHI, Dramatics & Theatre Society, IISER Kolkata *2025 – Present*
- **Active Member** – Slashdot, Game Development & Coding Club, IISER Kolkata *2025 – Present*

Online Profiles

- **Portfolio:** <https://berasankhadeep20-lang.github.io>
- **GitHub:** <https://github.com/berasankhadeep20-lang>
- **Codeforces:** https://codeforces.com/profile/Ronnie_Deep_04
- **LinkedIn:** <https://www.linkedin.com/in/sankhadeep-bera-4a49b436b/>